

Personalized nutrition: fruit and vegetables consumption in obesity prevention

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2.1 Introduction

Overweight and obesity are defined by the World Health Organization (WHO) as a condition in which the percentage of body fat increases to the extent that health and well-being are impaired. Given the prevalence of obesity, the WHO has declared that it has reached epidemic proportions worldwide (Lorenzo et al., 2016).

Among the causes of overweight and obesity, the energy imbalance between calories consumed and calories expended stands out mainly (Obesidad, 2021). The multiple factors lead to the complex interaction of genetic factors, alteration in energy expenditure, and increased oxidative stress in tissues, which produce the accumulation of proinflammatory cytokines, which regulate the metabolism of adipocytes and glucose and favor the development of pathologies such as cardiovascular diseases, hypertension, type 2 diabetes mellitus, hyperlipidemia, cerebrovascular accidents, certain cancers (endometrial, breast, ovarian, prostate, liver, gallbladder, kidney, and colon) (González Jiménez, 2011; Guh et al., 2009; Lorenzo et al., 2016). In addition, arterial hypertension and type 2 diabetes mellitus contribute to sedentary behaviors related to obesity itself as an endless spiral between cause and effect, which limits health and decreases the quality of life of patients (Guh et al., 2009). Nutrition is the basis for preventing or treating all the aforementioned complications.

Food provides nutrients (carbohydrates, proteins, fats, vitamins, and minerals) to the cell to maintain its vital functions. Food, especially of plant origin, also contains bioactive components; substances obtained from secondary metabolites that, although not considered essential for human health, have a significant impact and may be important in avoiding oxidative stress and maintaining long-term good health. (Martínez-Navarrete et al., 2008).

Oxidative stress, inflammatory pathways, and the expression of certain genes involved in the pathogenesis of chronic diseases are reduced by the intake of plant-based foods due to their fiber content, essential micronutrients, and bioactive components. Among the latter, polyphenols, plant sterols, xanthines, and extracts of medicinal plants stand out (Martínez-Navarrete et al., 2008).

The genotype can be homozygous, which is characterized by having two identical alleles on both homologous chromosomes, or heterozygous, when these alleles are different (Membrana nuclear, 2021). Variations in the DNA sequence between individuals are known as polymorphisms. This variability is responsible for the characteristics of everyone. Polymorphisms can be found in the coding regions of the genome or in non-coding regions, but they may have a regulatory or structural function. They can be homozygous or heterozygous. When polymorphisms affect a single nucleotide, they are called single-nucleotide polymorphisms (SNPs). SNPs do not generally cause diseases, but they can be associated with susceptibility to different pathological processes, so some SNPs are considered markers of interest for the diagnosis or study of the molecular mechanisms of a disease (Torrades, 2002). The SNPs in protein genes involved in the absorption, availability, metabolism, and excretion of nutrients and bioactive compounds contribute to the interindividual variability of their physiological mechanisms, daily requirements, and metabolic aspects that may intervene in the health and disease of the patient. The phenotypic effect does not depend on a single polymorphism but on a group of them (Borel & Desmarchelier, 2018).

Genome-wide association studies (GWAS) and candidate gene studies (CGAS) have identified associations between DNA variants and the physiological response to diet and its components, as, for example, affected functional pathways can lead to changes in the plasmatic concentration of certain nutrients or bioavailability (de Roos, 2013).

When we talk about bioavailability, we refer to the fraction and speed at which an active compound reaches the therapeutic target (Research, 2019). After ingesting a food, the digestion process begins in the mouth, then the food bolus passes to the esophagus and stomach, where it mixes with gastric juices and goes to the small intestine, where the absorption of the bioactive components occurs into the blood and through the portal vein until it reaches the liver (Enterocito, 2021).

Bioavailability varies depending on the category of bioactive compounds, their food source, and the way the food is processed prior to consumption (Milenkovic et al., 2017). The individual's age, sex, eating habits, their microbiota, and their genetic polymorphisms also affect the bioavailability of a bioactive compound, since allele variants can be related to low or high enzyme activity and thus achieve an effect individually as a response to each bioactive compound (Milenkovic et al., 2017).

Based on the objective of the study, it was proposed to prepare a matrix of recommendations on the consumption of dietary bioactives of potential interest in the control of obesity and its comorbidities. The hypothesis based on the GWAS and CGAS studies states that everyone has a SNP and can respond individually to bioactive components of the diet. Giving interest and importance to the study of genetics allows us to provide more personalized dietary recommendations for everyone.

2.2 Methodology

This narrative bibliographic review was carried out by constructing and applying specific search equations in the PubMed database, as indicated in the final master's thesis (Abad, 2021). A bibliographic search was carried out through the construction and application of two specific search equations. After the first search equation "dietary bioactive [Title/Abstract] AND obesity [Title/Abstract]" a list of components that are related to beneficial effects on obesity and its complications was obtained. Subsequently, the second search equation was applied in the same scientific database with "component name [Title/Abstract] AND obesity [Title/Abstract] AND genetic polymorphism."

The inclusion criteria for the selection of articles were a source of publication in the PubMed scientific database until June 2021; in English, as it is the main language used in scientific journals, the MeSH terms were used (Abad, 2021). In the second search, studies on the genetic response to dietary bioactive components in the control of obesity in humans were included. Studies that were not conducted on humans were excluded.

The identified articles were filed in the "Zotero" bibliographic manager to maintain cohesion in the bibliographic citations. In the selection of the articles, the inclusion criteria were applied during the reading of the abstract of the article or the complete article. The potential bioactive components were obtained, which were registered in a Microsoft Excel matrix, which allowed evaluating the risk of information bias. All the studies show and analyze the correlation between the gene variant in obesity and its comorbidities, diet, and indicators of nutritional status. The alleles of each gene were identified using the Gene Cards, SNP data base, and SNPedia databases (Rs1801282, 2021; Snpedia).

This review focuses on the inter-individual variability regarding the results of obesity complications and how they respond to bioactive components to summarize the information and develop a matrix of nutrigenetic recommendations that can serve health professionals and nutrition as a guide to dietary recommendations for patients. The food sources of the bioactive components were obtained from the FoodDB database and from the Vitonica information page (Gottau, 2012).

2.3 Results and discussion

The bibliographic search allowed us to find 33 bioactive components that are related to obesity; however, only nine of them correlate with the two search equations. Table 2.1 includes the results of anthocyanins in BMI and central obesity (Kalt et al., 2020; Lee et al., 2017), quercetin and its relationship with cardiovascular disease (Carrasco-Pozo et al., 2019; Chen et al., 2016b), and epigallocatechin gallate (EGCG) and its effect on reducing body weight. (Carrasco-Pozo et al., 2019; Chen, Liu et al., 2016; Vázquez Cisneros et al., 2017), plant sterols and LDL cholesterol levels (Glenn et al., 2021; Stelmach-Mardas et al., 2015), Bofutsushosan as a possible adjunct to treatment in diabetes mellitus 2 (Kuo et al., 2019; Shimada et al., 2008), apple juice consumption and its impact on the lipid profile (Habanova, et al., 2022; Mattes & Campbell, 2009), Caffeine and the metabolic changes it produces and therefore can be recommended in obese patients (Hursel & Westerterp-Plantenga, 2013; Westerterp-Plantenga et al., 2005), Capsaicin and its

TABLE 2.1 Bioactive components of food and their relationship with obesity and its comorbidities.

Component	Outcome	References
Anthocyanins	Dietary anthocyanins may be a potential regulator of inflammation derived from obesity; therefore, their consumption could be recommended in patients with a high BMI.	Lee et al. (2017)
Anthocyanins	The consumption of anthocyanins is related to a decrease in central obesity in women.	Kalt et al. (2020)
Quercetin	Quercetin induced apoptosis of fat cells.	Chen et al. (2016a)
Quercetin	Quercetin consumption seems to reduce the risk of cardiovascular diseases in patients with obesity.	Carrasco-Pozo et al. (2019)
EGCG	The consumption of EGCG seems to reduce the risk of cardiovascular diseases in patients with obesity.	Carrasco-Pozo et al. (2019)
EGCG	Consumption of green tea with 100–460 mg of EGCG per day is related to a decrease in body weight.	Vazquez Cisneros et al. (2017)
EGCG	The consumption of 856.8 mg/day of EGCG is related to the decrease in body weight and plasmatic levels of cholesterol and LDL.	Chen et al. (2016a)
Plant sterols	Maintaining a diet rich in plant sterols decreases the risk of coronary heart disease.	Glenn et al. (2021)
Plant sterols	The consumption of plant sterols is related to a decrease in LDL cholesterol in obese women (the data are not significant).	Stelmach-Mardas et al. (2015)
Bofutsushosan	The baicalin in Bofutsushosan has an activating effect on ISR-1 and GLUT4, so it could be used in the treatment and management of diabetes mellitus.	Kuo et al. (2019)
Bofutsushosan	Bofutsushosan treatment was associated with decreased visceral fat accumulation in mice.	Shimada et al. (2008)
Apple juice	The consumption of 300 mL/day of apple juice is related to an improvement in the lipid profile (LDL/HDL) in overweight women.	Habanova et al. (2022)
Apple juice	Consumption of an applesauce snack is associated with a stronger appetizer response than liquid apple juice in obese patients.	Mattes and Campbell (2009)
Caffeine	Caffeine consumption was associated with weight loss in women with obesity.	Westerterp-Plantenga et al. (2005)
Caffeine	The consumption of tea rich in catechins and caffeine has thermogenic effects and can produce metabolic changes.	Hursel et al. (2013)
Capsaicin	Capsaicin treatment in obesity reduces body weight.	Zheng et al. (2017)
Capsaicin	The consumption of capsinoids seems to increase the activity of brown adipose tissue and therefore contribute to a decrease in body fat.	Yoneshiro et al. (2013)
Resveratrol	Resveratrol 150 mg/day supplementation was associated with AMPK activation in obese humans.	Timmers et al. (2011)
Resveratrol	Resveratrol snack consumption linked to obesity benefits.	Ahmad and Gani (2021)
Resveratrol	Resveratrol treatment linked to improved insulin secretion and sensitivity in mice.	Abo Alrob et al. (2022)

contribution in reducing body fat ([Yoneshiro et al., 2013](#); [Zheng et al., 2017](#)) and the Resveratrol and its benefit in obesity ([Ahmad & Gani, 2021](#); [Alrob et al., 2022](#); [Timmers et al., 2011](#)).

Subsequently, the second search equation was performed. Subsequently, the second search equation was performed. We included 36 articles that evidence 25 interactions of eight types of bioactives or extracts with 20 SNPs in 14 genes ([Fig. 2.1](#)). The influence of the genetic variants identified in the response to each family of bioactives considered in obesity and its comorbidities is summarized in [Tables 2.2–2.5](#).

2.3.1 Influence of genetic variants on the response to flavonoids in obesity

In the family of flavonoids, differential responses of parameters related to metabolic health to anthocyanin intake have been reported depending on different polymorphisms (SNPs) in the gene for the PON1 enzyme. A deficit in PON1 has

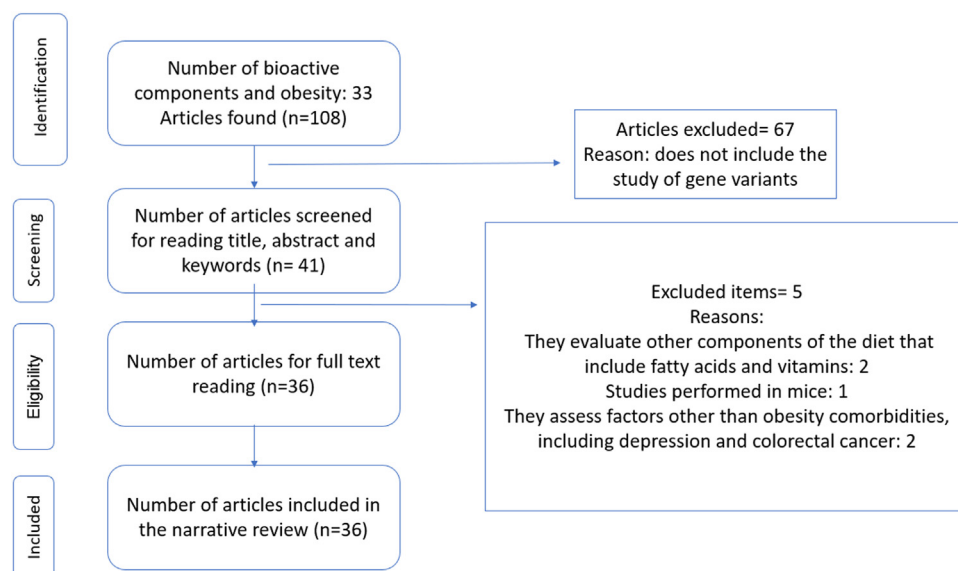


FIGURE 2.1 PRISMA. Subsequently, the second search equation was performed. We included 36 articles that evidence 25 interactions of 8 types of bioactives or extracts with 20 SNPs in 14 genes.

been associated with oxidative stress and cardiovascular disease. PON1 participates in the protection of HDL oxidation, and the polymorphisms found respond positively to a high intake of anthocyanins, which contribute to the protection of HDL (Kim et al., 2017; Rizzi et al., 2016). The intake of anthocyanins increases HDL cholesterol levels; the increase of 1 mg of HDL in the blood reduces the risk of coronary artery disease by 2% in men and 3% in women, so from a nutritional point of view, it can recommend the consumption of anthocyanins in obese patients with A, T, or CC alleles of the PON1 gene. For some alleles, anthocyanins have an opposite effect on the normal physiology of the PON1 enzyme. In this case, the intake of this component should be controlled, without neglecting the consumption of unsaturated fatty acids that help maintain HDL levels (Table 2.2) (Rizzi et al., 2016).

Quercetin could have a protective role against cardiovascular diseases, depending on the genotype or phenotype of apolipoprotein E (apoE). ApoE is found in different circulating lipoproteins and plays an essential role in the catabolism of triglyceride molecules bound to them and the maintenance of vascular health (Egert et al., 2010). Quercetin intake was associated with a significant reduction in systolic blood pressure for the apoE3 phenotype (classification for people with the E3/E3 genotype). A reduction of ~3 mmHg is associated with a 5% reduction in congenital heart disease (CHD) mortality. However, for apoE4 carriers, quercetin proved to have an opposite function since the ratio of LDL: HDL cholesterol increased in this phenotype (Table 2.2) (Egert et al., 2010). The apoE3/2 genotype is associated with higher total cholesterol levels, lower than the apoE4/4 and apoE4/3 genotypes after caffeine intake, which means that for these last two genotypes, the intake of 2.9 cups of coffee per day can alter the lipid profile of obese patients due to a decrease in normal apolipoprotein function. For the apoE3/2 genotype, caffeine shows a protective role. It should be mentioned that in this study, they did not differentiate between normal and decaffeinated coffee types. Therefore, additional studies are required to analyze this relationship (Table 2.3). The nutritional recommendation on the intake of caffeine and quercetin should be based on the apolipoprotein phenotype of each patient to prevent obesity comorbidities, such as arterial hypertension and coronary events (Milenkovic et al., 2017).

Another enzyme related to the variation in blood pressure is catechol methyltransferase (COMT), an enzyme that inactivates neurotransmitters such as adrenaline, a signal that, when released, can increase heart rate and raise blood pressure (EL COLESTEROL FORMA ÁCIDOS BILIARES. s. f., 2021). The consumption of EGCG could be included in the nutritional recommendations for AA homozygotes for the SNP rs4680 of the COMT gene since it demonstrates a protective effect on systolic blood pressure. Special care must be taken in carriers of the G allele since it can be counterproductive and lead to hyperinsulinemia (Table 2.2) (Milenkovic et al., 2017).

2.3.2 Influence of genetic variants on the response to plant sterols in obesity

The intake of plant sterols is related to an improvement in the lipid profile in patients with obesity, as it is related to the NPC1L1 protein found in the cells of the gastrointestinal tract and participates in regulating cholesterol absorption. An important limitation of this study is the exclusion of women. The polymorphisms found in NPC1L1 are haplotypes, that

TABLE 2.2 Influence of genetic variants on the response to flavonoids in obesity.

Bioactive	SNP	Gene	Gene name	Genotype/ Fenotype	Type o study	Number of participants	Sex	Age	Poblation	Main result	References
Anthocyanins	Rs854549	PON1	Paraoxonase 1	CC versus AA	CGAS	500	Both	20–85	Italy	A high intake of anthocyanins (from 25.7 to 614.4 mg/day) is related to increased HDL levels in CC homozygotes, decreased in AA homozygotes, and unchanged in heterozygotes	Rizzi et al. (2016)
	Rs854551	PON1	Paraoxonase 1	A > G	CGAS	500	Both	20–85	Italy	The intake of 25.7 to 614.4 mg/day of anthocyanins is related to a decrease in the atherogenic index in plasma (0.07) in carriers of the A allele. Homozygous GG without significant changes	
	Rs854552	PON1	Paraoxonase 1	C > T	CGAS	500	Both	20–85	Italy	Intake of 25.7 to 614.4 mg/day of anthocyanins is associated with increased HDL levels (5.6 mg/dL) in carriers of the C allele. T allele associated with lower HDL levels (2 mg/dL)	

(Continued)

TABLE 2.2 (Continued)

Bioactive	SNP	Gene	Gene name	Genotype/ Fenotype	Type o study	Number of participants	Sex	Age	Poblation	Main result	References
Rs854571	PON1		Paraoxonase 1	T > C	CGAS	500	Both	20–85	Italy	The intake of 25.7 to 614.4 mg/day of anthocyanins is related to higher HDL levels (3.92 mg/dL) in carriers of the T allele. C allele is not related to altered HDL levels	Egert et al. (2010)
Rs854572	PON1		Paraoxonase 1	C > G	CGAS	500	Both	20–85	Italy	The intake of 25.7 to 614.4 mg/day of anthocyanins is related to increased HDL-C levels (3.94 mg/dL) in carriers of the C allele. GG homozygotes are associated with lower HDL-C levels (2 mm/dL)	
Quercetin	Rs429358	APOE	Apolipoprotein	E3/3 versus E4/4	Experimental study	93	Both	25–65	Germany	ApoE3 carriers showed a 3.4 mmHg reduction in systolic blood pressure after 6 weeks of treatment with 150 mg/d of quercetin, in apoE4 carriers no significant reductions in blood pressure were found	

	Rs429358	APOE	Apolipoprotein	E4/4 versus. E3/ 3	Experimental study	93	Both	25–65	Germany	ApoE4 carriers showed a reduction in HDL and apoA1 and an increase in the LDL:HDL ratio after 6 weeks of treatment with 150 mg/d of quercetin, in apoE3 carriers no significant reductions in HDL-C were found	
EGCG	Rs4680	COMT	Catechol-O-methyltransferase	AA versus GG	Experimental study	20	Both	18–70	United Kingdom	AA homozygotes show a more significant reduction than GG in blood pressure after ingesting 836 mg of EGCG (by 6.2 mmHg systolic and 7.7 mmHg diastolic)	Milenkovic et al. (2017)
	Rs4680	COMT	Catechol-O-methyltransferase	AA versus GG	Experimental study	20	Both	18–70	United Kingdom	GG heterozygotes show 50%–70% higher plasma insulin levels after 120 min of ingesting 836 mg of EGCG, there is no difference in plasma glucose levels	
	Rs4680	COMT	Catechol-O-methyltransferase	AA versus GG	Experimental study	50	Both	major a 18	European descent	AA homozygotes show lower blood pressure than GG homozygotes after ingesting 1.06 g of green tea extract	Milenkovic et al. (2017)

TABLE 2.3 Influence of genetic variants on the response to other bioactive components in obesity.

Bioactive	SNP	Gene	Gene name	Genotype/ Fenotype	Type o study	Number of participants	Sex	Age	Poblation	Main result	Reference
Caffeine	Rs7412	APOE	Apolipoprotein	E3/2 versus E4/4 + E4/3	GWAS	76069	Both	50–72	Denmark	ApoE32 carriers were associated with lower cholesterol levels (10%) and lower caffeine intake (approx 2.4 tz coffee per day); E44 + E43 were related to a coffee intake of around 2.9 cups per day and higher cholesterol levels	Williams et al. (1985)
	Rs7412	APOE	Apolipoprotein	E2 /E3 /E4	Experimental study	120	Both	30–65	Sweden	ApoE2 carriers have been shown to have lower cholesterol levels; however, studies are lacking to determine whether the intake of 600 mL of coffee per day for 4 weeks has any effect	Strandhagen et al. (2004b)
	Rs1801133	MTHFR	Methylenetetrahydrofolatereductase	CC > TT	Experimental study	120	Both	29–65	Europe	TT homozygote carriers show higher homocysteine levels than CC homozygotes, after ingesting 600 mL of coffee per day	Strandhagen et al. (2004b)

TABLE 2.3 (Continued)

Bioactive	SNP	Gene	Gene name	Genotype/ Fenotype	Type o study	Number of participants	Sex	Age	Poblation	Main result	Reference
	Rs659366	UCP2	Uncoupling protein 2	CC > TT	Experimental study	9	Men	20–30	Japan	Supplementation with 150 mg capsaicin/day has no effect on the cardiac autonomic nervous system in obese patients who perform physical exercise.	Shin et al. (2008)

TABLE 2.4 Influence of genetic variants on the response to plant sterols in obesity.

Bioactive	SNP	Gene	Gene name	Genotype/ Fenotipe	Type o study	Number of participants	Sex	Age	Poblation	Main result	References
Plant sterols	Rs2072183	NPC1L1	Niemann-Pick C1-Like 1	C > G; Leu272	Experimental study	82	Men	Mayor a 18	European descent	Carriers of the G allele show a 2.4fold reduction in LDL cholesterol after ingesting 2 g of plant sterols per day for 4 weeks. Haplotype with rs523613 G > A	Milenkovic et al. (2017); Zhao et al. (2008)
	Rs3808607	CYP7A1	Colesterol 7 α -hidroxilase	A > C; Low > High activity	Experimental study	67	Both	20–75	European descent	Carriers of the C allele present a greater reduction in total cholesterol compared to AA homozygotes, after supplementation with 2 g of plant sterols	Milenkovic et al. (2017)
	Rs3808607	CYP7A1	Colesterol 7 α -hidroxilase	A > C	Experimental study	67	Both	20–75	European descent	Carriers of the C allele present greater reduction in total cholesterol and increase in plasma lanosterol levels, after supplementation with 3.2 g of plant sterols	De Castro- Orós et al. (2011)
	Rs4148217	ABCG8	ATP binding cassette	C > A; Thr40> 0Lys	Experimental study	82	Men	Mayor a 18	European descent	Carriers of the A allele show a 3.9fold greater reduction in LDL cholesterol than carriers of the C allele after ingesting 2 g of plant sterols per day for 4 weeks	Milenkovic et al. (2017); Zhao et al. (2008)

(Continued)

TABLE 2.4 (Continued)

Bioactive	SNP	Gene	Gene name	Genotype/ Fenotype	Type o study	Number of participants	Sex	Age	Poblation	Main result	References
	Rs523613	NPC1L1	Niemann-Pick C1-Like 1	G > A; Arg406 > Ter	Experimental study	82	Men	Mayor a 18	European descent	Carriers of the A allele show a 2.4fold reduction in LDL cholesterol after ingesting 2 g of plant sterols per day for 4 weeks. Haplotype with rs2072183 C > G	
	Rs6720173	ABCG5	ATP binding cassette	T > C	Experimental study	74	Both	40–70	Finland	Carriers of the C allele show higher blood latosterol/ cholestanol ratios	Gylling et al. (2007)
	Rs4148217	ABCG8	ATP binding cassette	A > C	Experimental study	74	Both	40–70	Finland	Carriers of the C allele show higher blood cholesterol/ campesterol ratios	
	Rs429358	APOE	Apolipoprotein	E2 versus E3/ E4	Experimental study	207	Both	21–75	Chicago	ApoE2 carriers show a better response in the lipid profile in relation to apoE3 and apoE4, after ingesting 1.1–2.2 g plant sterols/day	Sanchez- Muniz et al. (2009)

TABLE 2.5 Influence of genetic variants on the response to extracts in obesity.

Bioactive	SNP	Gene	Gene name	Genotype/ Fenotype	Type o study	Number of participants	Sex	Age	Poblation	Main result	References
Bofutsushosan	Rs4994	ADRB3	B3 adrenergic receptor	TT versus CC; Trp64 > Arg	Experimental study	166	Both	18–65	Korea	TT homozygotes show a more significant reduction in body weight, BMI, waist circumference, percentage body fat, and diastolic pressure. Carriers of the C allele is related to a decrease in HDL-C after taking Bofutsushosan	Park et al. (2014)
	Rs5443	GNB3	Guanine nucleotide binding protein beta-3 subunit	C > T; Ser275 > Ser	Experimental study	166	Both	18–65	Korea	C allele carriers show greater decrease in body weight, BMI, body fat after ingestion of 237 mg Bofutsushosan 12 times daily for 8 weeks	
	Rs659366	UCP2	Uncoupling protein 2	DD versus. DI + II; AA versus GA + GG	Experimental study	166	Both	18–65	Korea	DD homozygotes show a reduction in body weight, body fat mass, total cholesterol, HDLc after treatment with 237 mg/d of Bofutsushosan	

(Continued)

TABLE 2.5 (Continued)

Bioactive	SNP	Gene	Gene name	Genotype/ Fenotype	Type o study	Number of participants	Sex	Age	Poblation	Main result	References
										extract. Carriers of the I allele is not related to significant changes	
	Rs659366	UCP2	Uncoupling protein 2	DI + II versus DD; AG + AA versus GG	Experimental study	166	Both	18–65	Korea	Carriers of the A allele are associated with a more significant reduction in systolic blood pressure than DD homozygotes after taking Bofutsushosan	
	Rs1801282	PPARY2	Peroxisome proliferator- activated receptor gamma	CC versus CG; Pro/Pro versus Pro/ Ala; Pro12Ala	Experimental study	166	Both	18–65	Korea	CC homozygotes demonstrated a significant reduction in body weight, BMI, waist circumference, HDL-C, body fat percentage. Heterozygotes were associated with a significant increase in HDL-C after taking Bofutsushosan	

Apple juice	Rs1800795	IL6	Interleucin 6	G > C	Experimental study	68	Men	Older than 18	European descent	Carriers of the C allele are related to a decrease in body fat after ingesting 750 mL of apple juice (rich in polyphenols 802.5 mg/day). CG and GG allele carriers did not show a significant reduction in body fat	Barth et al. (2012)
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is, combinations of SNPs that occur together, while if the patient has the rs523613 G > A polymorphism, they will also have the rs2072183 C > G polymorphism, so the intake of plant sterols would maintain the reduction of LDL in both polymorphisms (Zhao et al., 2008, n.d.). The protein encoded by the ABCG8 gene also acts at the intestinal level, promoting the excretion of cholesterol and facilitating the transport of sterols back to the lumen. It is seen that the change of alleles C > A contributes to a better response of the gene and a decrease in cholesterol LDL (Table 2.4) (Gylling et al., 2007).

The cholesterol 7- α -hydroxylase enzyme, which participates in the biosynthesis of bile acids, also responds positively to plant sterols, generating an increase in intestinal bile acids and a decrease in blood cholesterol levels (Table 2.4) (Caimari et al., 2010; De Castro-Orós et al., 2011; El, 2021).

Apolipoprotein 2 carriers show a better response in the lipid profile after ingesting plant sterols (Sanchez-Muniz et al., 2009).

2.3.3 Influence of genetic variants on the response to plant extracts in obesity

Table 2.5 summarizes the responses to the extracts of medicinal plants. The Bofutsushosan extract could also be included as part of nutritional recommendations due to its association with the protein that encodes the ADRB3 gene, a receptor located in adipose tissue that regulates lipolysis and thermogenesis (Receptor de adenosina A2a ADORA2A Homo sapiens humano—Gene—NCBI. s. f., 2021). The extract is also related to the GNB3 gene, which encodes receptors and signal transduction effectors related to adipose tissue (n.d.). The protein encoded by the PPAR γ gene is a regulator of adipocyte differentiation and activates the PON1 gene, which increases the synthesis and release of the paraoxonase enzyme in the liver, reducing the risk of atherosclerosis (PubChem. s. f., 2021). The nutritional recommendation of the Bofutsushosan extract should be made depending on the gene allele; its daily intake is related to the reduction of body weight, BMI, and body fat, which is associated with preventive and treatment recommendations in the management of obesity. Although gastrointestinal problems are known adverse effects of the consumption of this extract, the authors of the study assure that there was no toxic impact on the liver or kidney function of the participants (Table 2.5) (Park et al., 2014).

UCP2, a protein that facilitates anion transfer between mitochondrial membranes, is especially expressed in skeletal muscle and plays an important role in thermogenesis (Gen l, 2021). A response association of this protein with the Bofutsushosan extract and capsinoids has been identified by promoting the reduction of abdominal fat (Snitker et al., 2009).

2.3.4 Influence of genetic variants on the response to other compounds in obesity

Capsaicin has also been shown to be related to TRPV1, the capsinoid receptor that is activated by increases in temperature commonly caused by the ingestion of spicy foods (TRPV1—Transient Receptor Potential Cation Channel Subfamily V Member 1 (Human), n.d.). The intake of capsinoids has a cardio-protective effect since the decrease in abdominal fat would mean a decrease in waist circumference and a decrease in the risk of cardiovascular disease (Table 2.3) (Snitker et al., 2009). Capsaicin supplementation has no effect on the autonomic nervous system in patients with obesity (Shin & Moritani, 2008).

Regarding caffeine intake, there is a response by the alpha 2B adrenergic receptor that regulates the release of adrenaline and norepinephrine and its corresponding effect on blood pressure. The C allele does not seem to show a response in the release of these hormones after caffeine intake (adrenoceptor alpha 2B Homo sapiens human—Gene—NCBI. s. f., 2021). The protein encoded by the ADORA2A gene plays a role in heart rate, circulation, and renal blood flow. After caffeine ingestion, TT homozygotes have been shown to have increased activity of this gene (Table 2.3) (Receptor de Adenosina A2a ADORA2A, n.d.). Although coffee intake was also associated with apoE2 individuals, who demonstrated a protective effect from increased cholesterol, more research is needed as the data is unclear (Strandhagen et al., 2004a). The increase in homocysteine levels is related to damage to the arteries and increased risk of clots; it is also related to TT homozygotes after coffee intake (Strandhagen, Zetterberg, Aires, Palmér, Rymo, Blennow, Landaas, et al., 2004b).

Interleukin 6 is produced at sites of inflammation, while the presence of obesity and type 2 diabetes mellitus increases its levels (Gen l, 2021). Although it has been shown to have a positive response to the presence of polyphenols, they can decrease body fat percentage significantly, indicating that for obese patients with the C allele of the IL6 gene, apple juice intake can be included in the nutritional recommendations for these patients (Table 2.3) (Barth et al., 2012).

All these results obtained from the narrative bibliographic review have allowed the creation of a proposal for dietary recommendations based on the identified polymorphisms, determining their limitations and nutritional applications (Table 2.6). The matrix is made up of four columns: the SNP related to your gene, the genotype/phenotype/allele that responds specifically to each component, and the nutrigenetic advice.

The WHO advises consuming 400 g of fruits and vegetables daily as part of a healthy diet to prevent chronic diseases such as cardiovascular disease, cancer, type 2 diabetes, and obesity. However, today, nutrigenetics allows us to investigate how individual genetic differences influence how a person responds to their diet and how these genetic factors can impact their health. The data obtained in this narrative review allow us to personalize recommendations on the consumption of fruits and vegetables. This means that, within the recommended 400 g, we can identify which specific fruits have the most beneficial effect on a person's health due to their bioactive components and the individual genetics of each patient (World Health Organization, 2015).

According to data from the National Health and Nutrition Survey (ENSANUT) in Ecuador, the average consumption of fruits and vegetables is insufficient and does not meet international recommendations, reaching only 183 g/day on average. Among the adult population, men between 31 and 50 years old have the highest average consumption, with 212 g/day, while men between 51 and 59 years old have the lowest consumption, with 171 g/day. These data are worrying given that Ecuador has a wide variety of fruits and vegetables available, but the population does not seem to understand the importance of their consumption (Ministerio de Salud Pública del Ecuador, 2014).

Therefore, the implementation of public strategies and programs aimed at motivating the population to consume more fruits and vegetables is suggested, with the aim of preventing chronic non-communicable diseases. Furthermore, it would be relevant to propose a genetic study of the population, since, as mentioned in Table 2.6, dietary recommendations should be based on personalized nutrition. Some foods with previously investigated bioactive components are listed in Table 2.7 (Gambini et al., 2013).

The results of this narrative bibliographical review suggest recommending the intake of bioactive components based on gene variants. The study of gene polymorphisms is becoming an essential tool in the recommendations of nutrition and diet plans. It is known that individual genetic variability defines the efficacy of bioactive components in health, which has led to the use of different molecular techniques for the identification of markers for the prevention and treatment of obesity and its comorbidities, as well as for personalization of the recommendations provided during the consultation by health and nutrition professionals (Ayala & Emilia, 2007).

Gene-diet interaction analysis studies have allowed us to collect important data on the different genotypes and phenotypes and how they respond to the different bioactive components, such as the rs659366 polymorphism of the UCP2 gene associated with the development of obesity, has been positively regulated by the intake of capsinoids, with benefits in lipid oxidation and the loss of adipose tissue at the abdominal level (Fur et al., 2004).

These results could indicate that the intake of antioxidants has a significant impact and is essential to maintaining good health in the long term (Martínez-Navarrete et al., 2008). However, there are genetic polymorphisms that respond negatively to some bioactive components. Such is the case of apoE4, associated with a reduction in HDL levels and an increase in the LDL:HDL ratio after treatment with quercetin. Egert et al. suggest that the data should be confirmed with prospective studies where the apoE genotype is assessed as a determinant of the responsiveness to quercetin (Egert et al., 2010). In addition, the amount of this bioactive component used in the study is hardly covered by food, so special care should be taken with the consumption of quercetin supplements, but not with the intake of food sources. Another relevant example is the carriers of homozygous TT alleles in the rs2236624 polymorphism of the adenosine receptor, for whom the intake of 3 mg of caffeine/kg of weight can increase systolic blood pressure. The limitations of the study include the small sample size and the fact that the target population was non-smoking men, so the data could not be extrapolated to the entire population (Renda et al., 2012). However, considering eating habits and statistical data that report that 63% of the adult Spanish population drinks an average of 3.6 cups of coffee, these data can serve as a reference to prevent arterial hypertension and for the development of study comparisons in Ecuador (FoodData Central, s. f., 2020).

Thus, there are also bioactive components such as Bofutsushosan that can develop adverse effects such as gastrointestinal discomfort in obese patients, although Park et al. mentioned that more studies are needed to elucidate whether the genetic polymorphisms analyzed contribute to this adverse response (Park et al., 2014). In the case of the UCP2 and PPARY2 polymorphisms, the reduction in body weight, total cholesterol, and HDL cholesterol stands out; therefore, the nutrigenetic recommendations in Table 2.6 include the recommendation to encourage the consumption of foods rich in unsaturated fatty acids and balance the effects of the bioactive component (Park et al., 2014).

The information recorded in this study allowed the preparation of a matrix with personalized recommendations for the treatment and prevention of obesity and its comorbidities. Food nutrients are vital for cell function; however,

TABLE 2.6 Recommendations on the consumption of bioactive compounds.

Bioactive	SNP	Gene	Genotype/ fenotype/ allele	Nutritional recommendations
Anthocyanins	Rs854549	PON1	CC	The intake of foods rich in anthocyanins such as blueberries, figs, soybeans and grapes is recommended to cover the recommendation of 25.7–614.4 mg/day, in order to optimize HDL levels
	Rs854551	PON1	A carriers	
	Rs854552	PON1	C carriers	
	Rs854571	PON1	T carriers	
	Rs854572	PON1	C carriers	
	Rs854549	PON1	AA	Control the intake of anthocyanins, consume less than 25.7–614.4 mg/day, to avoid a decrease in HDL levels
	Rs854552	PON1	T carriers	
	Rs854572	PON1	GG	
Quercetin	Rs429358	APOE	E3/3	Increase the intake of foods rich in quercetin such as garlic, garden tomato and green pepper, to improve systolic blood pressure levels. Although to reach the recommendation of 150 mg/day, large amounts of these foods are necessary
	Rs429358	APOE	E4/4	Control the intake of foods rich in quercetin, do not consume capsules with more than 150 mg of bioactive per day. In order to avoid the increase in LDL levels
EGCG	Rs4680	COMT	AA	Include the intake of 836–1006 mg/day of EGCG, which is found in approximately 3 cups of green tea. In order to improve blood pressure levels
	Rs4680	COMT	GG	Avoid the intake of high amounts of EGCG, do not exceed two cups of green or black tea to avoid high levels of insulin in the blood
Plant sterol	Rs2072183	NPC1L1	G carriers	Include the intake of 2 g of plant sterols per day, contained in 200 mL of corn oil, sunflower oil and soybean oil, in order to improve total and LDL cholesterol levels
	Rs3808607	CYP7A1	C carriers	
	Rs4148217	ABCG8	A carriers	
	Rs523613	NPC1L1	A carriers	
Bofutsushosan	Rs4994	ADRB3	TT	Include the intake of dry Bofutsushosan extract, to improve body weight, BMI and body fat. Consider that there may be gastrointestinal discomfort
	Rs5443	GNB3	C carriers	
	Rs659366	UCP2	A carriers	
	Rs1801282	PPARY2	G carriers	
	Rs659366	UCP2	DD	Include the intake of dry Bofutsushosan extract, to improve total cholesterol levels and body fat mass. Keep in mind that there may be gastrointestinal discomfort. Consider eating foods rich in unsaturated fatty acids such as walnuts, almonds, and olive oil to avoid lowering HDLc
	Rs1801282	PPARY2	CC	
	Rs4994	ADRB3	CC, CT	Control high intake of Bofutsushosan to avoid a reduction in HDL levels
Apple juice	Rs1800795	IL6	C carriers	Include in the daily diet the intake of 750 mL of apple juice rich in polyphenols, as well as foods rich in polyphenols such as blackberries, celery stalks, tomatoes to obtain a reduction in body fat
Caffeine	Rs4680	COMT	G carriers	Maintain the consumption of 3 mg of caffeine per Kg of weight per day, for 2 days to avoid the increase in blood pressure levels
	Rs2229169	ADRA2B	C carriers	

(Continued)

TABLE 2.6 (Continued)

Bioactive	SNP	Gene	Genotype/ fenotipe/ allele	Nutritional recommendations
Caffeine	Rs7412	APOE	E4/4	Control caffeine intake, it is recommended to consume less than 2.9 cups of coffee per day to avoid an increase in total cholesterol levels
	Rs7412	APOE	E4/3 carriers	
	Rs7412	APOE	E3/2 carriers	Caffeine intake is recommended, 2.4 cups of coffee per day in order to optimize total cholesterol levels
	Rs2236624	ANDORA2A	TT	Control caffeine intake, it is recommended to consume less than 3 mg of caffeine per kg of patient per day, in order to avoid an increase in systolic blood pressure. As a comparison, a cup of espresso contains 80 mg of caffeine
	Rs2229169	ADRA2B	A carriers	
Capsinoid	Rs222741	TRPV1	Val carriers	The intake of 6 mg of capsinoids per day, found in green, orange and red peppers, as well as in pepper and chili, is recommended in order to reduce abdominal fat
	Rs659366	UCP2	A carriers	

TABLE 2.7 Food sources of bioactive compounds.

Bioactive compound	Fruit and vegetables
Anthocyanins	Blueberry, cherry, raspberry
Quercetin	Onion, apple, blueberry, raspberry
Plant sterols	Spinach, carrot, cabbage
Capsaicin	Paprika, pepper, chili
Resveratrol	Blackberry, blueberry, grape

bioactive compounds have a significant impact and are complementary to dietary recommendations that can be provided in an individualized consultation, with effective and permanent results (Martínez-Navarrete et al., 2008). Health professional consultations last twenty minutes on average, which is insufficient time for personalized attention and reading of gene interactions. For this reason, the application of the matrix with nutrigenetic recommendations is suggested as a useful tool for calculating nutritional plans for outpatients and hospitalized patients. Likewise, for the diagnosis of nutritional status, the analysis of gene expression related to food intake should be considered a fundamental pillar to knowing the pathophysiological status of a patient, since genes are not the direct cause of diseases, but they do predispose to them (Calvar, 2015).

There are other determinants of lifestyle, such as alcohol consumption, smoking, emotions, stress, physical activity, hours of sleep, and drugs that could also change the expression of some genes. This indicates that multidisciplinary teamwork is important for the treatment and prevention of obesity and its comorbidities (Calvar, 2015). Likewise, it is considered important to provide the patient with a set of nutritional recommendations that include reducing salt intake, including fish oil, and dietary patterns based on the DASH or Mediterranean diet, which contain a variety of fruits, vegetables, and rich foods in fiber for a comprehensive treatment (Appel et al., 2006; Ayala & Emilia, 2007).

2.4 Conclusions

After the bibliographic review carried out in the PubMed database, eight bioactive compounds were found that demonstrated evidence of effects on obesity in clinical studies and that were compared with genetic polymorphisms. The

results obtained are presented in tables, in the form of summaries detailing the bioactive compound, the SNP, gene, genotype, type of study, number of participants, sex and age of the individuals, and the main result in a specific description. The response and comparison of the interaction of each allele with the intake of the bioactive compound are described, as well as the difference in response in homozygous and heterozygous individuals. Additionally, a study was identified that associates two haplotypes with a decrease in LDL levels.

The analysis and interpretation of the information collected allowed the elaboration of a matrix of recommendations based on the identified polymorphisms. The application of this matrix is suggested as a useful tool in the integral evaluation of the nutritional status of the patient.

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